Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

Mid Semester Test (MST)

BE- 3rd Year, 25-26 ODDSEM	Course Code: UCS520
Date: Sept 30, 2025	Course Name: Computer Network
M. Marks: 40 (Weightage:30)	Name of Faculty: SHD, GVR, AMR, SSH, MCL

Note: Attempt all questions. Attempt all parts of one question at one place. Assume missing data if any.

Q. 1 (a)	Given the data word 1010011110 and the divisor 10111, i. Show the generation of the codeword at the sender site using CRC. ii. Suppose the second bit from the left is inverted during transmission. How will receiver detect this error?	CO2,L3 (5)
(b)	Find the Checksum value of 1001001110010011 and 1001100001001101 of 16-bit segment at the sender site. Also show the checking of the codeword at the receiver site.	CO2,L1 (3)
(c)	Why does the CSMA/CD algorithm fail in wireless LANs (IEEE 802.11)? Briefly discuss the hidden and exposed problems	CO1,L2 (2)
Q. 2 (a)	Create a system of three LANs with four bridges. The bridges (B1 to B4) connect the LANs as follows: • B1 connects LAN1 and LAN2 • B2 connects LAN1 and LAN3 • B3 connects LAN2 and LAN3 • B4 connects LAN1, LAN2 and LAN3 Choose B1 as the root bridge. Show the forwarding and blocking ports after applying the spanning tree procedure.	CO2,L1 (4)
(b)	i. A non-periodic composite signal has a bandwidth of 400 kHz, with a middle frequency of 240 kHz and peak amplitude of 30 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal. ii. A noiseless channel of 3 kHz bandwidth transmits signals using 8 levels. Find the maximum data rate.	CO1,L2 (3)
(c)	Illustrate the frame structure for standard Ethernet (IEEE 802.3). For each field in the frame, specify its size in bytes and describe its primary function in the frame encapsulation and delivery process.	CO1,L2 (3)
Q. 3 (a)	A sender uses a sliding window protocol with a window size of 4 and sequence numbers from 0 to 7. It begins by transmitting frames 0, 1, 2, and 3. Frame 1 is lost during transmission. Frame 3 arrives at the receiver before Frame 2, and both arrive before the missing Frame 1 is retransmitted. The receiver operates under two different rules: ▪ Go-Back-N (GBN): Sends a positive acknowledgment (ACK) only for in-order frames. ▪ Selective Repeat (SR): Sends an ACK for each correctly received frame and a negative acknowledgment (NAK) for any lost frame. **Based on this scenario, answer the following:** i. Describe the sequence of events (what is sent, received, and acknowledged) for both the GBN and SR protocols with suitable diagram. ii. Identify the specific frames the sender must retransmit in each case. iii. Explain why the SR protocol is more efficient than Go-Back-N in this scenario.	CO2,L1 (4)
(b)	Write a short note on following with suitable example: i. Polling and Reservation mechanisms ii. Router and Switch Devices	CO1,L3 (4)

(c)	An ALOHA network transmits 400-bit frames over a 400-kbps channel. i. For a Slotted ALOHA system with an offered load of 2000 frames per second, what is the throughput (in frames per second)? ii. For a Pure ALOHA system with an offered load of 500 frames per second, what is the throughput (in frames per second)?	CO1,L2 (2)
Q. 4 (a)	Draw the graph of the data stream 00110011 using each of the following schemes, assuming that the last signal level has been positive. i. NRZ-L scheme ii. NRZ-I scheme iii. Polar RZ scheme iv. Manchester and v. Differential Manchester scheme	CO2,L3 (5)
(b)	A device is sending out data at the rate of 1000 bps. i. How long does it take to send out 10 bits? ii. How long does it take to send out a single character?	CO1,L1 (2)
(c)	For each of the following networks, discuss the consequences if a connection fails, and how does the degree of recovery time in each case? i. Five devices arranged in a star topology (not counting the hub) ii. Five devices arranged in a bus topology iii. Five devices arranged in a ring topology	CO1,L3 (3)
	********** All the Best **********	